RELATIVE PSEUDO-COHERENCE OF COMPLEXES IS INVARIANT UNDER PUSHFORWARD ALONG FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16994-17001 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17003-17013, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Cited as an input by this same wave's chart card L16132; the translation carries explicit side conditions, not a naked cite.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma relative-pseudo-coherence-characterize (the affine-local test for relative pseudo-coherence); More on Algebra, lemma finite-extension-pseudo-coherent; Derived Categories of Schemes, lemma quasi-coherence-direct-image.

Source reference. Original lemma 09UJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16837–16847. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0673: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17013–17013. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08D5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17013–17013. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes locally of finite type over a base scheme S. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Then E is m-pseudo-coherent relative to S if and only if $R\pi_*E$ is m-pseudo-coherent relative to S.*

Proof. Translation of the result of More on Algebra, Lemma 0673 into the language of schemes. Observe that $R\pi_*$ indeed maps $D_{QCoh}(\mathcal{O}_X)$ into $D_{QCoh}(\mathcal{O}_Y)$ by

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Derived Categories of Schemes, Lemma 08D5. To do the translation use Lemma 09UJ. □